

Satellite Structural Inspection Interferometric Synthetic Aperture Radar (InSAR) Based

INDEX

Sr. No.	Description	Page No.
1.	Methodology	1
	1 Satellite SAR System with Resolution and Revisit Time	
	2 Retrieving Multi-Temporal SAR Imagery Over the Specified Period	
	3 Differential SAR Interferometry (D-InSAR) for Displacement Measurement	2
2.	Visualization	3
	1 Defining specific bridge location & parameters for satellite SAR based inspection	
	2 Calibration, georeferencing, orthorectification of SAR images (VH/VV)	4
	3 Results of SAR based bridge inspection, including visualizations	5
	3.1 VV Displacement From 23 Jan 2025 to 28 June 2025	6
	3.2 VH Displacement From 23 Jan 2025 to 28 June 2025	7
	3.3 SAR Based Analysis at Abutment 1 LHS	8
	3.4 SAR Based Analysis at Abutment 1 RHS	9
	3.5 SAR Based Analysis at Pier 1 LHS	10
	3.6 SAR Based Analysis at Pier 1 RHS	11
	3.7 SAR Based Analysis at Abutment 2 LHS	12
	3.8 SAR Based Analysis at Abutment 2 RHS	13
3.	Analysis	14
	1 Identify potential deformation, shifts or movement of the bridge structure	
	Conclusion	

METHODOLOGY

1. Satellite SAR System with Resolution and Revisit Time

To select a suitable SAR system, technical specifications such as spatial resolution, frequency band, polarization, and revisit time were considered. Sentinel-1, operating at C-band (5.405 GHz), was selected due to:

Parameter	Sentinel-1 C-SAR (IW Mode)
Frequency Band	C-band (~5.405 GHz)
Revisit Time	12 days (single satellite)
Spatial Resolution	5 m (range) × 20 m (azimuth)
Coverage Mode	Interferometric Wide Swath (IW)

2. Retrieving Multi-Temporal SAR Imagery Over the Specified Period

Multi-temporal **Sentinel-1 SLC IW mode** data were downloaded based on:

- **Area of Interest (AOI):** A polygon covering the bridge and surrounding terrain.
- **Date range:** Multiple acquisitions were collected spanning several months to capture temporal variations.
- **Image acquisition conditions:** All images were selected with the same relative orbit, pass direction (ascending or descending), and polarization (typically VV/VH) to ensure consistency for interferometric analysis.

S1A_IW_SLC__1SDV_20250111T130240_20250111T130310_057397_0710AB_813D
S1A_IW_SLC__1SDV_20250123T130239_20250123T130309_057572_07179E_5431
S1A_IW_SLC__1SDV_20250204T130238_20250204T130308_057747_071E86_3FE6
S1A_IW_SLC__1SDV_20250216T130238_20250216T130308_057922_072592_C312
S1A_IW_SLC__1SDV_20250228T130238_20250228T130307_058097_072CB4_7A14
S1A_IW_SLC__1SDV_20250312T130238_20250312T130308_058272_0733CA_217F
S1A_IW_SLC__1SDV_20250324T130238_20250324T130308_058447_073AB5_B445
S1A_IW_SLC__1SDV_20250405T130238_20250405T130308_058622_0741CE_2B52
S1A_IW_SLC__1SDV_20250417T130238_20250417T130308_058797_0748F4_3FFD
S1A_IW_SLC__1SDV_20250429T130239_20250429T130309_058972_07501E_08D8
S1A_IW_SLC__1SDV_20250511T130238_20250511T130308_059147_0756BA_DE13
S1A_IW_SLC__1SDV_20250523T130238_20250523T130308_059322_075CCB_14B2
S1A_IW_SLC__1SDV_20250604T130237_20250604T130307_059497_0762D9_3AB5
S1A_IW_SLC__1SDV_20250616T130237_20250616T130306_059672_0768D1_F57D
S1A_IW_SLC__1SDV_20250628T130236_20250628T130306_059847_076EE3_6BFD

3. Differential SAR Interferometry (D-InSAR) for Displacement Measurement

D-InSAR analysis was performed to measure millimeter-scale deformation over time:



VISUALIZATIONS

1. Defining specific bridge location & parameters for satellite SAR based inspection



Project & Bridge Details

State – Rajasthan
Bridge No. – 903/2
Bridge Chainage – 917+650

Project - Chittorgarh to Kota
Zone – RJ 06
Structure Type – MNB

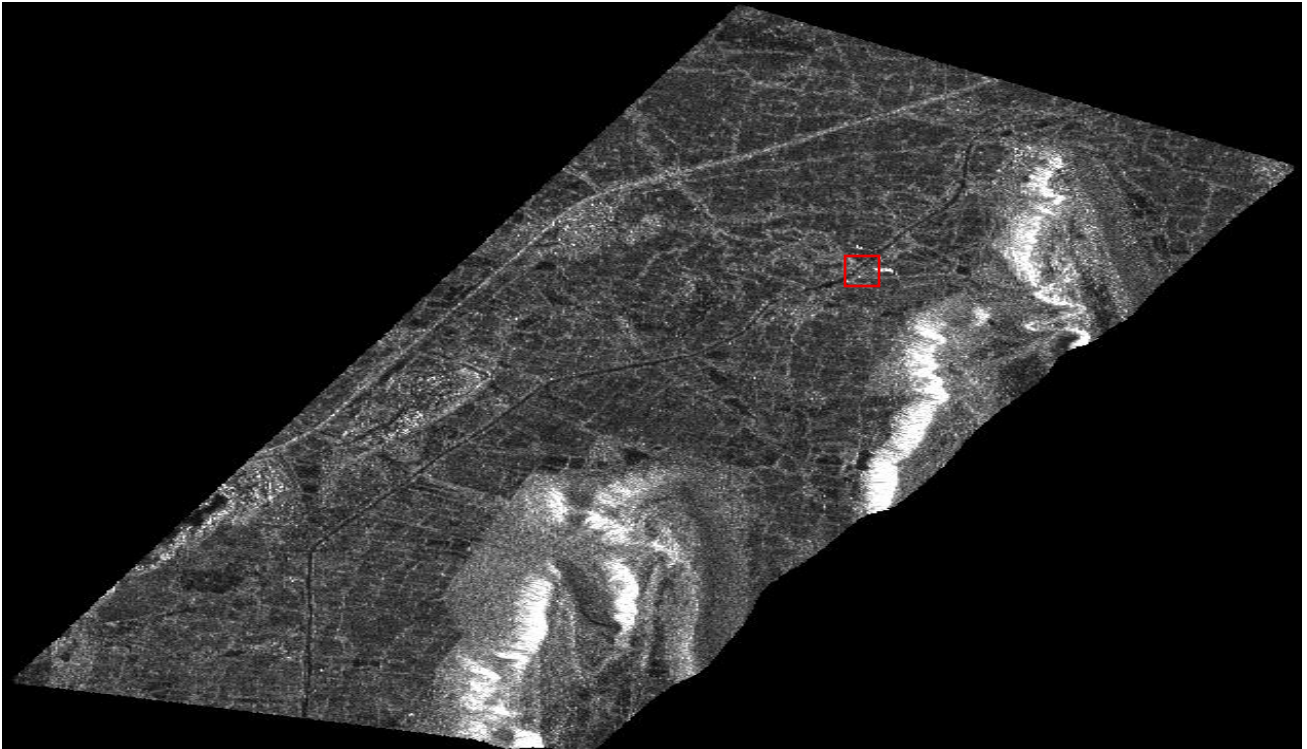
Sar Parameters

Sensor – Sentinel-1 (C-Band)
Beam Mode – IW (Interferometric Wide Swath)
Path – 34
Frame – 519

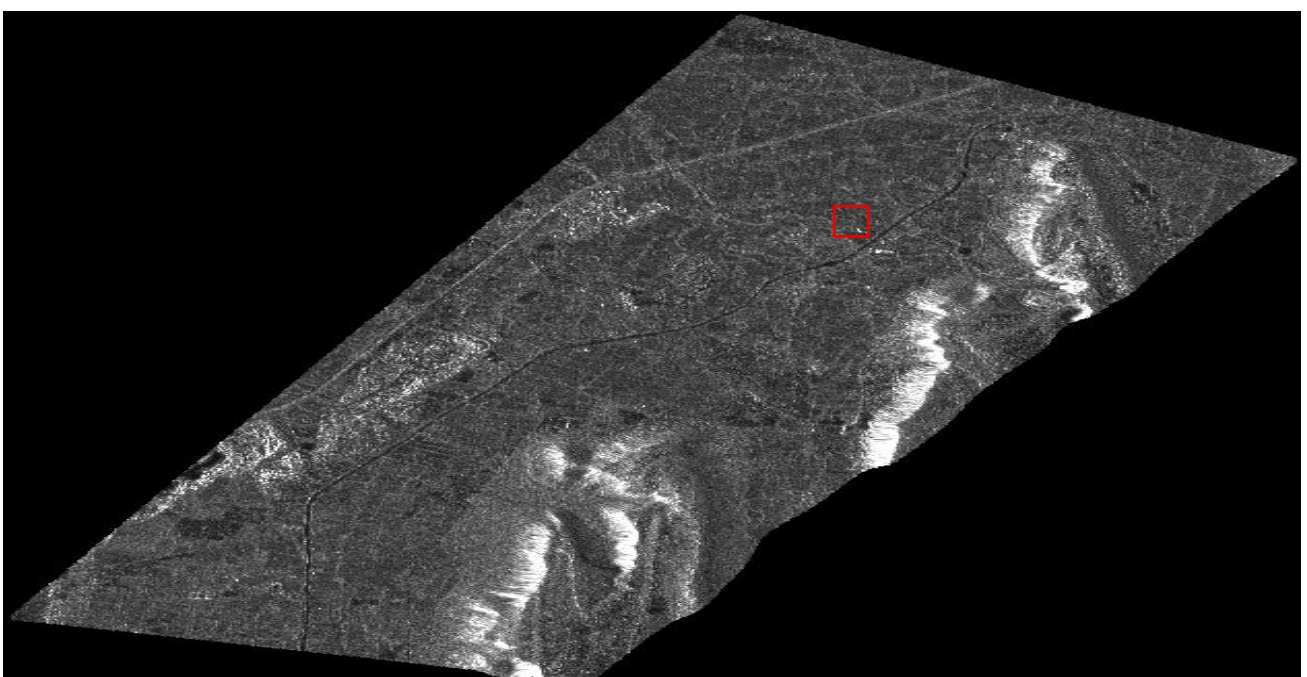
Flight Direction - DESCENDING
Polarization – VV+VH
Absolute Orbit – 54706

2. Calibration, georeferencing, orthorectification of SAR images (VH/VV)

- Orthorectification of SAR image VH Polarization



- Orthorectification of SAR image VV Polarization

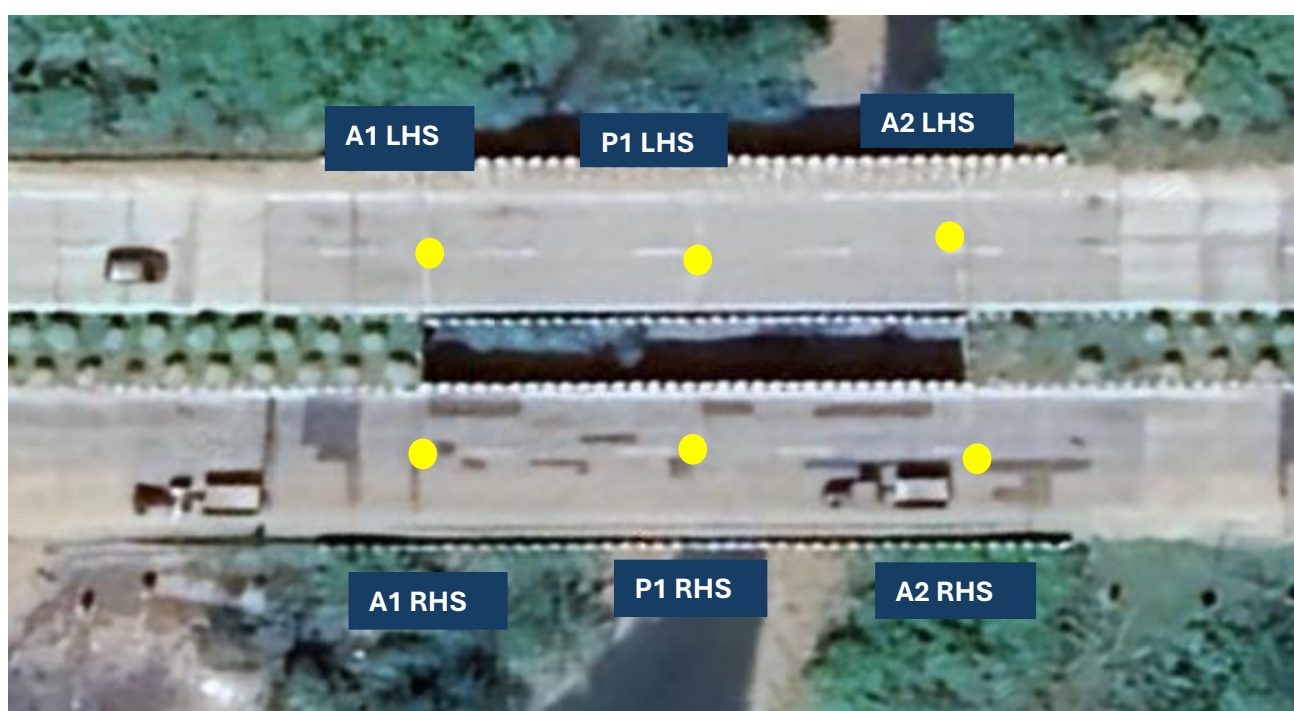


3. Results of SAR based bridge inspection, including visualizations

Bridge Components

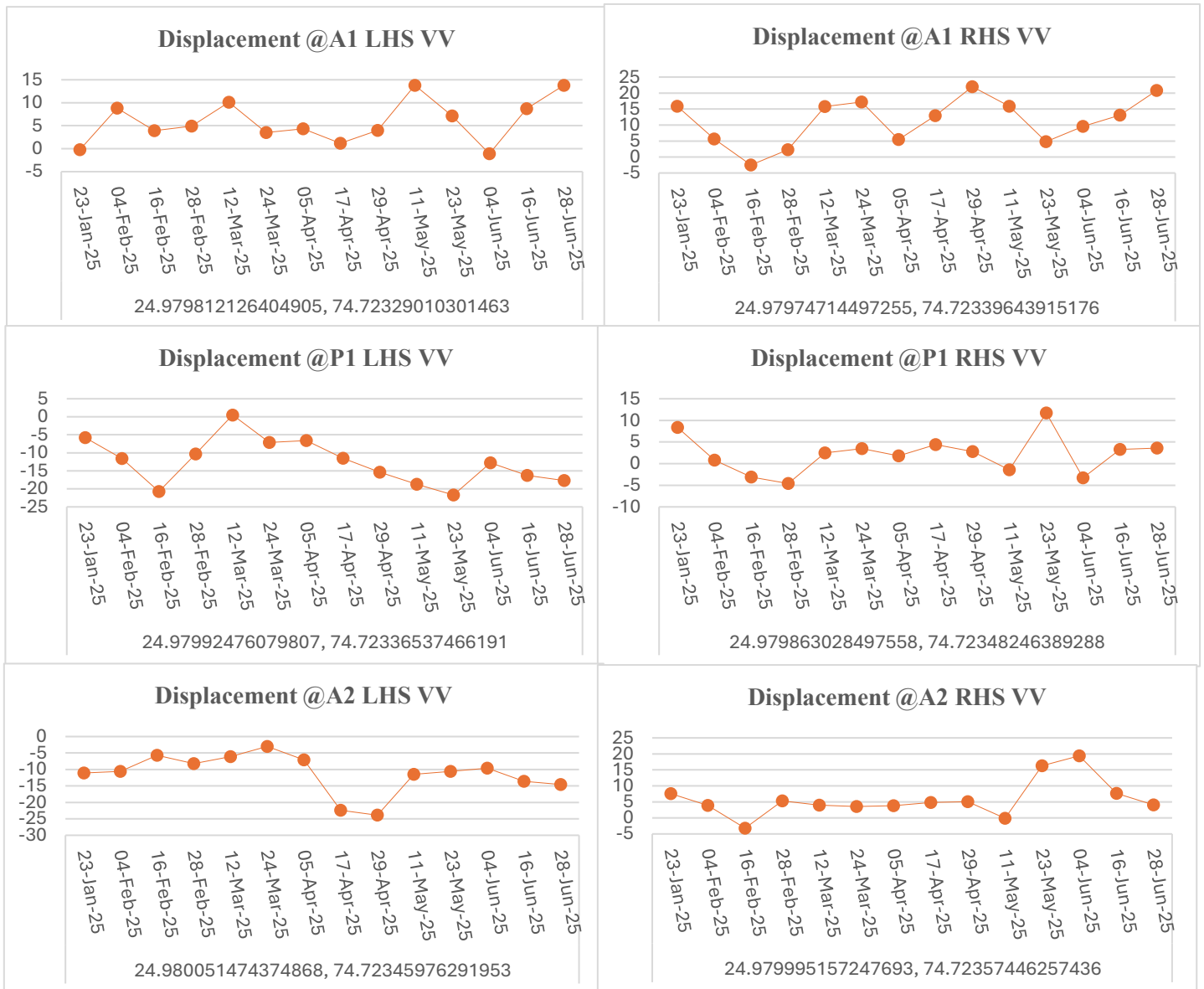
MNB – CH-917+650

Length – 31.73M



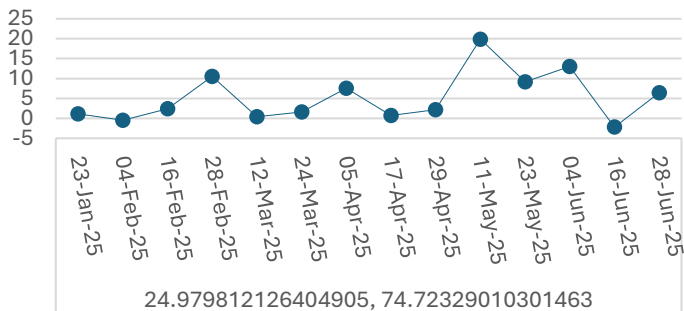
Scan Duration		From : 23 Jan 2025	To : 28 June 2025
Components & Abbreviation			
A1		Abutment 1	
P1		Pier 1	
A2		Abutment 2	
LHS		Left Hand Side	
RHS		Right Hand Side	

3.1 VV Displacement From 23 Jan 2025 To 28 June 2025

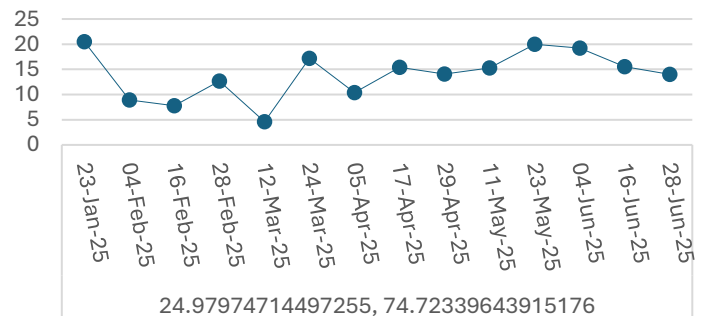


3.2 VH Displacement From 23 Jan 2025 To 28 June 2025

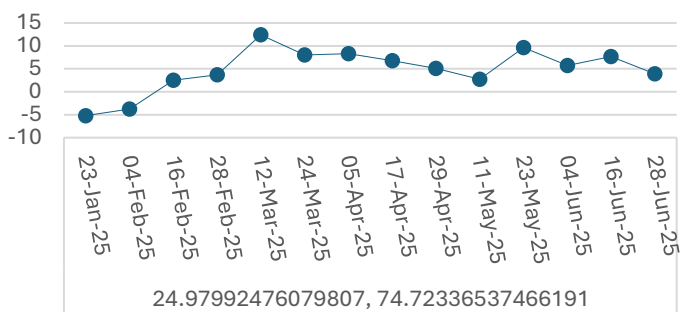
Displacement @A1 LHS VH



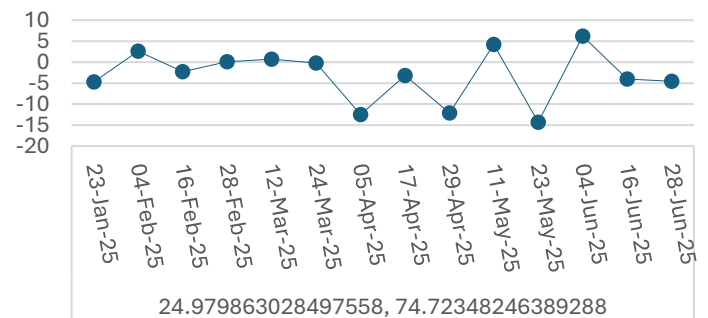
Displacement @A1 RHS VH



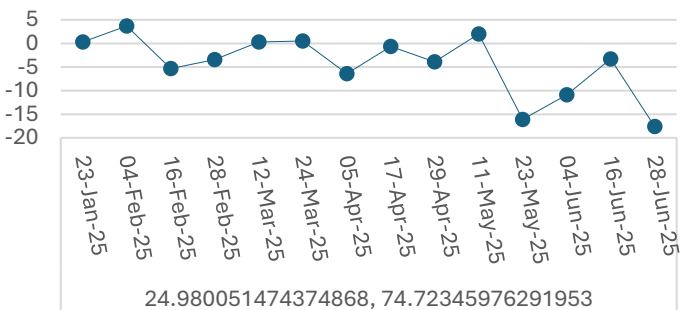
Displacement @P1 LHS VH



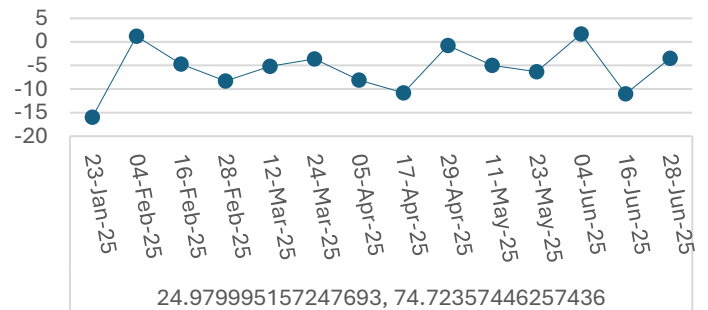
Displacement @P1 RHS VH



Displacement @A2 LHS VH



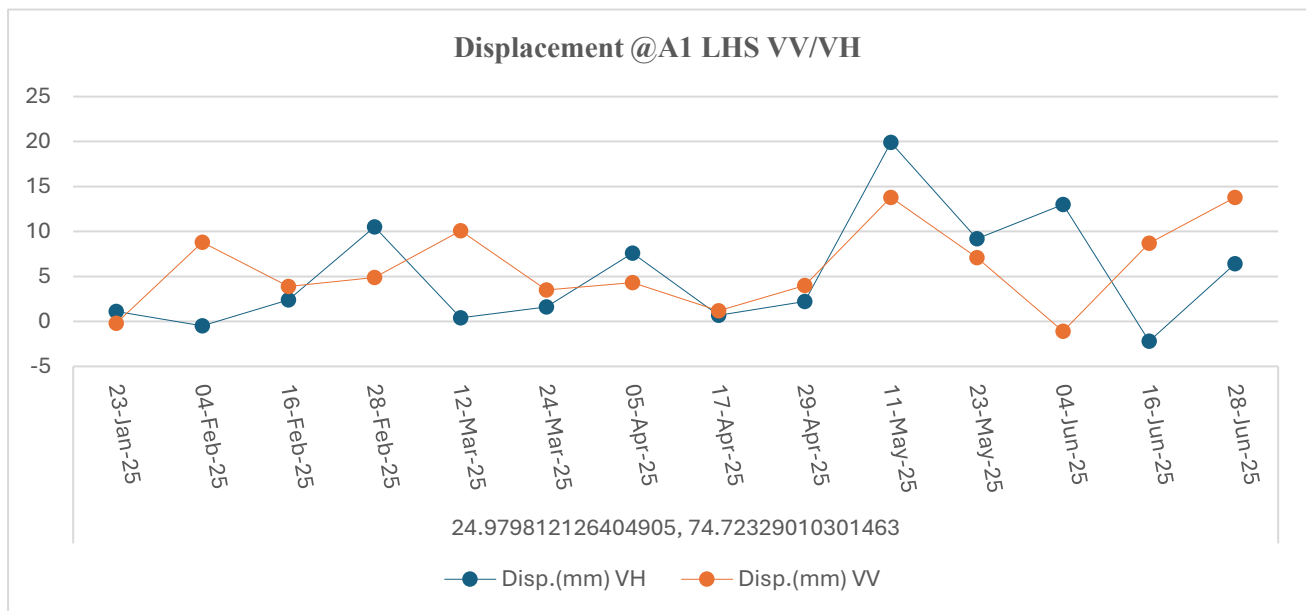
Displacement @A2 RHS VH



3.3 SAR Based Analysis at Abutment 1 LHS:



A1 LHS (24.979812126404905, 74.72329010301463)



Key Observations:

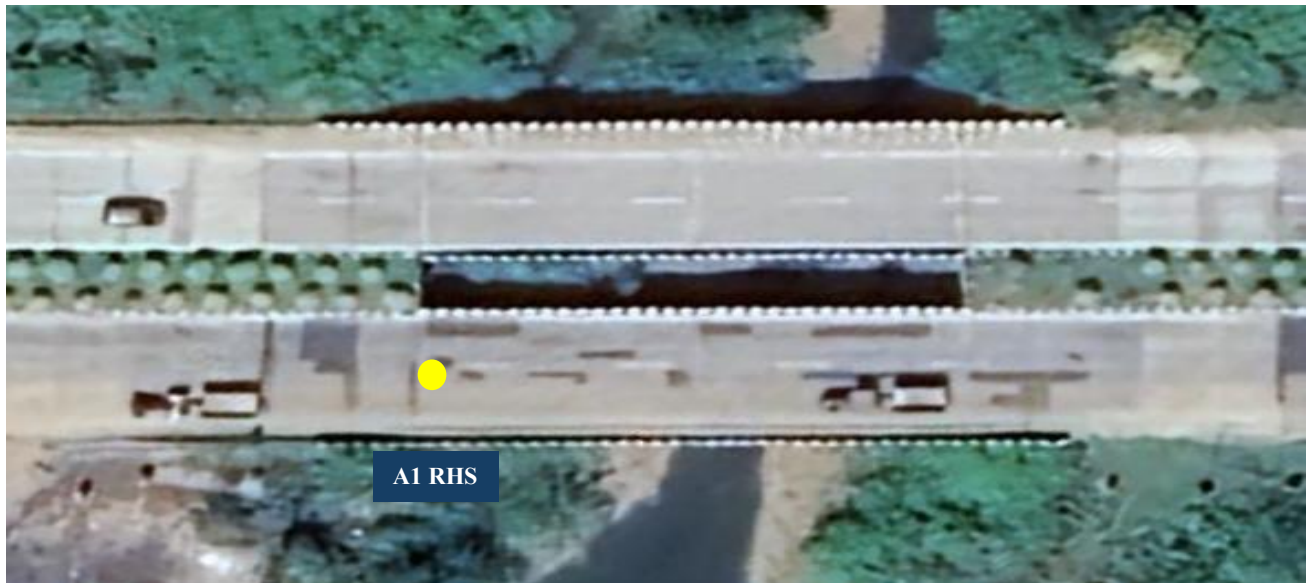
Displacement Magnitude Patterns:

- **Vertical Displacement (VV)** Vertical displacement exhibits moderate variability with an overall positive dominance during the monitoring period (Jan–Jun 2025), indicating net vertical uplift at the A1 LHS location.
- **Horizontal displacement (VH)** Horizontal displacement shows high variability with alternating uplift and settlement phases, ranging roughly between –3 mm and +20 mm.

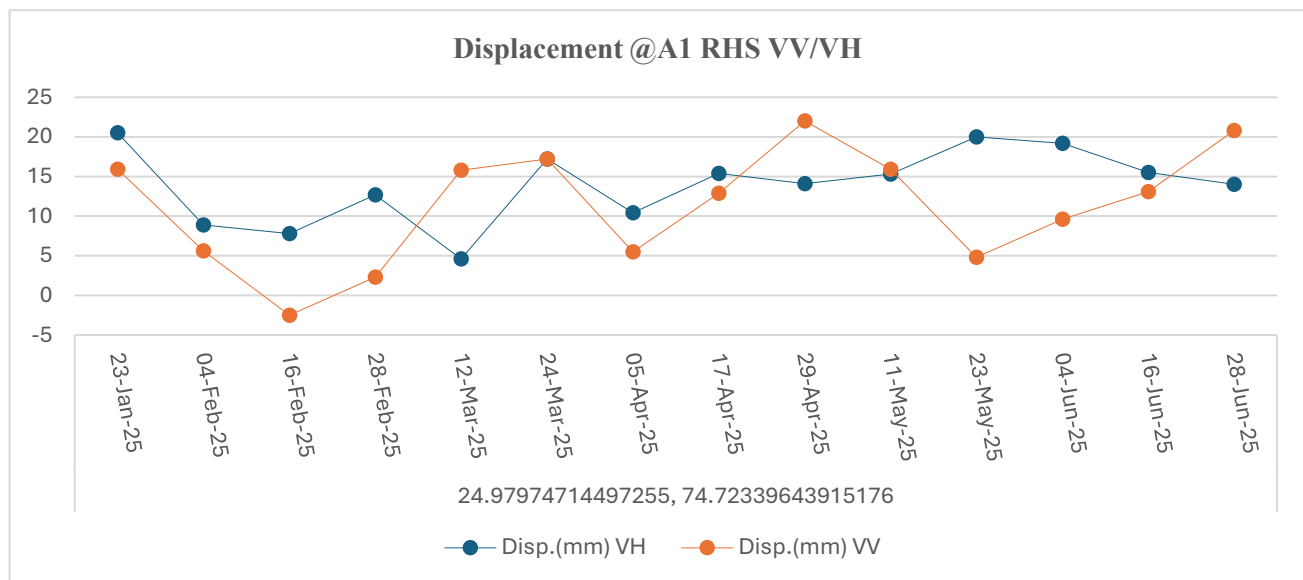
Critical Events & Anomalies:

- **January 2025:** VH begins near +1 mm, while VV is slightly positive (+2 mm), indicating a stable initial deformation state.
- **May 2025:** VH remains positive (+13 mm), while VV drops briefly to 0 mm, highlighting a decoupled response with lateral dominance.
- **June 2025:** VH recovers to +6 mm, while VV peaks again at +14 mm, confirming continued uplift-dominated vertical behaviour with ongoing lateral variability toward the end of the monitoring period.

3.4 SAR Based Analysis at Abutment 1 RHS:



A1 RHS (24.97974714497255, 74.72339643915176)



Key Observations:

Displacement Magnitude Patterns:

- **Horizontal displacement (VH)** Horizontal displacement shows moderate variability with a strong positive dominance, ranging roughly between +5 mm and +20 mm.
- **Vertical displacement (VV)** Vertical displacement exhibits moderate to high variability with a predominantly positive trend over the monitoring period, indicating overall vertical uplift at the A1 RHS location.

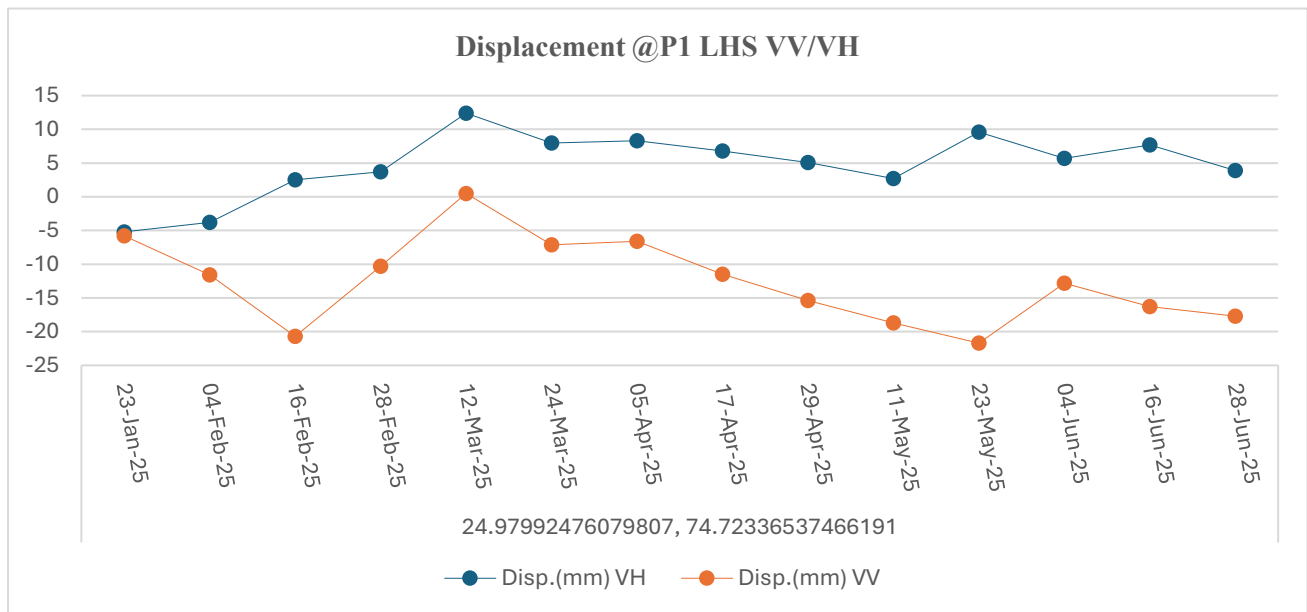
Critical Events & Anomalies:

- **January 2025:** VH starts high at +20 mm, while VV is also elevated (+15 mm), indicating a strong initial coupled uplift and lateral displacement phase.
- **March 2025:** VH drops to +5 mm, while VV rises sharply to +16 mm, highlighting a vertical-dominant uplift pulse.
- **June 2025:** VH stabilizes around +14 mm, while VV reaches +21 mm, confirming continued uplift-dominated vertical behaviour with sustained lateral displacement toward the end of the monitoring period.

3.5 SAR Based Analysis at Pier 1 LHS:



P1 LHS (24.97992476079807, 74.72336537466191)



Key Observations:

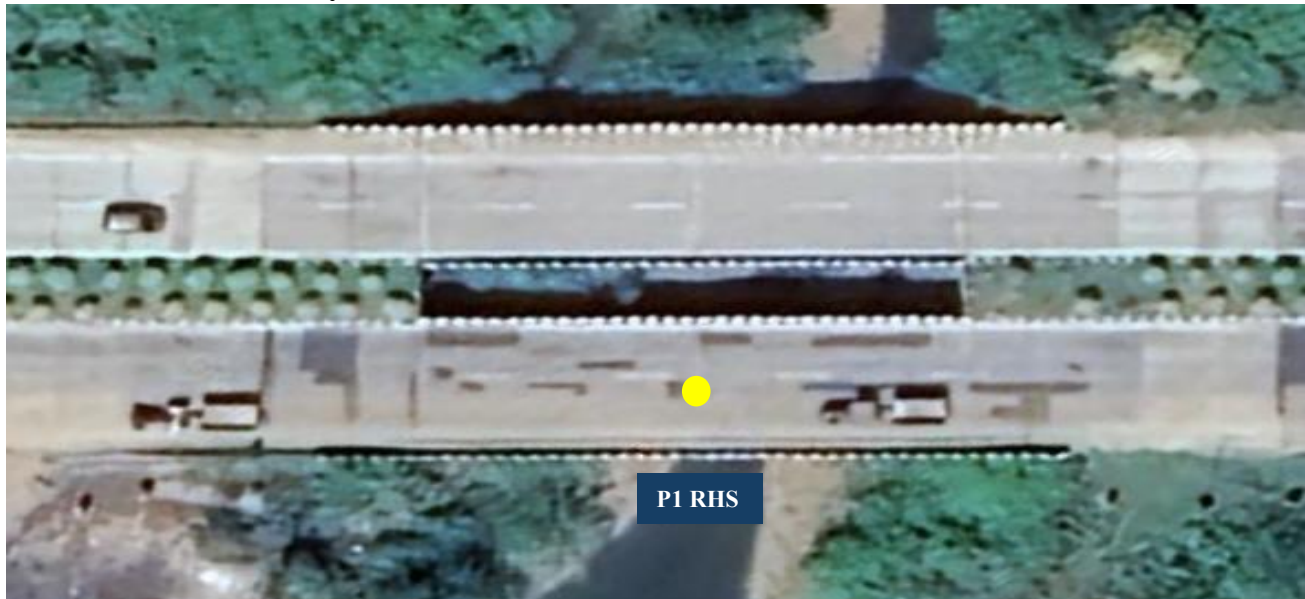
Displacement Magnitude Patterns:

- **Vertical Displacement (VV)** Vertical displacement exhibits a strongly negative and progressively intensifying trend over the monitoring period (Jan–Jun 2025), indicating persistent vertical settlement at the P1 LHS location.
- **Horizontal Displacement (VH)** Horizontal displacement shows moderate variability with a predominantly positive trend, ranging roughly between –4 mm and +12 mm.

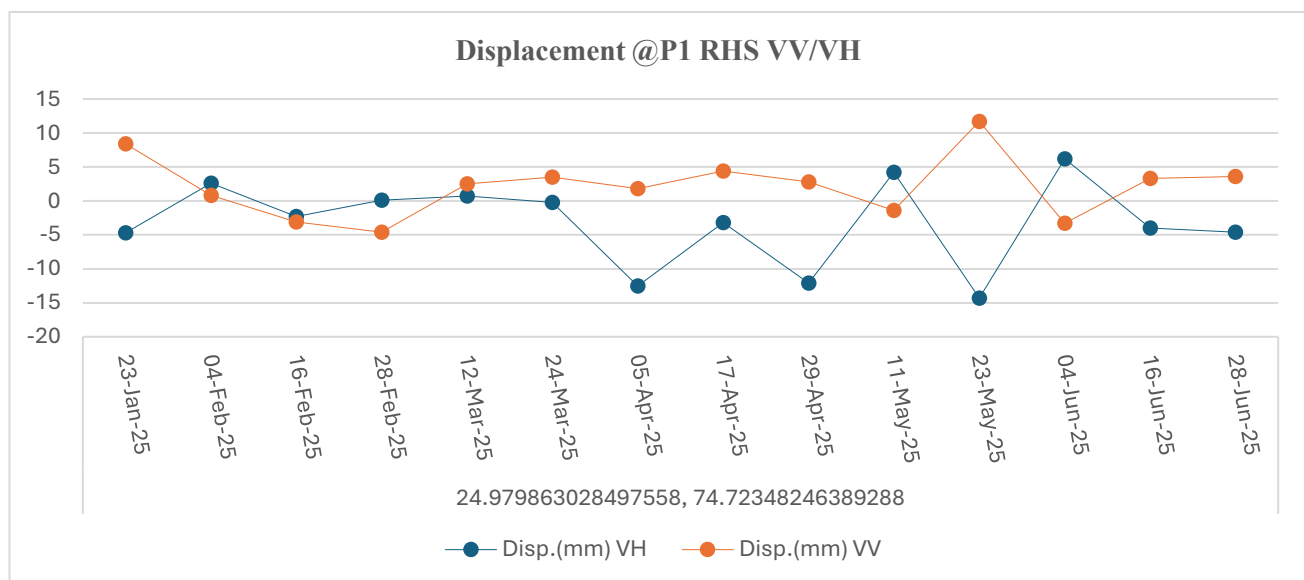
Critical Events & Anomalies:

- **January 2025:** VH begins near –4 mm, while VV is around –6 mm, indicating an initial coupled settlement phase affecting both displacement components.
- **May 2025:** VV reaches its most negative value (–22 mm), while VH increases to +9 mm, representing the most severe vertical settlement event combined with lateral displacement.
- **June 2025:** VH fluctuates between +4 and +8 mm, while VV remains strongly negative, confirming persistent settlement with no stabilization toward the end of the monitoring period.

3.6 SAR Based Analysis at Pier 1 RHS:



P1 RHS (24.979863028497558, 74.72348246389288)



Key Observations:

Displacement Magnitude Patterns:

- **Vertical Displacement (VV)** Vertical displacement exhibits moderate variability with alternating uplift and settlement phases over the monitoring period.
- **Horizontal Displacement (VH)** Horizontal displacement shows high variability with a strong negative dominance, ranging roughly between +6 mm and -15 mm.

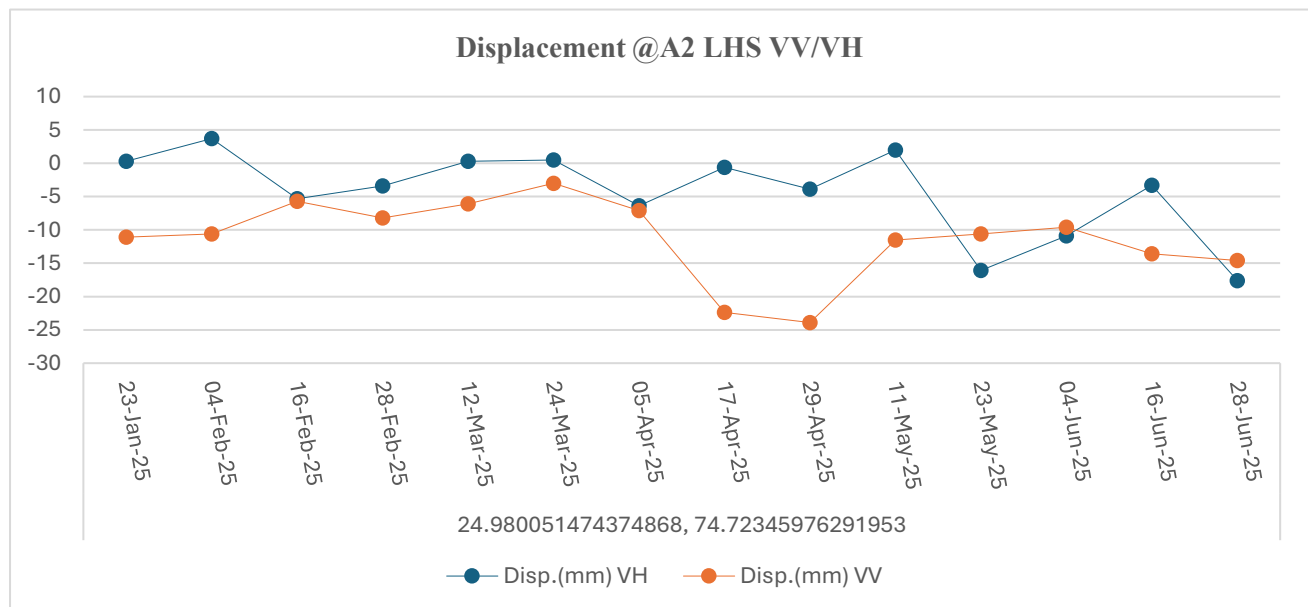
Critical Events & Anomalies:

- **January 2025:** VH starts near -5 mm, while VV is positive (+8 mm), indicating initial vertical uplift occurring alongside lateral settlement.
- **April 2025:** VH recovers briefly to -3 mm, then drops again to -12 mm, while VV fluctuates near +3 mm, indicating repeated lateral instability under relatively stable vertical behaviour.
- **June 2025:** VH declines again to -4 to -5 mm, while VV stabilises around +3-4 mm, confirming continued lateral variability with no vertical deterioration toward the end of the monitoring period.

3.9 SAR Based Analysis at Abutment 2 LHS:



A2 LHS (24.980051474374868, 74.72345976291953)



Key Observations:

Displacement Magnitude Patterns:

- **Vertical Displacement (VV)** Vertical displacement is predominantly negative throughout the monitoring period, indicating overall vertical settlement at the A2 LHS location.
- **Horizontal Displacement (VH)** Horizontal displacement exhibits high variability with alternating uplift and settlement phases, ranging roughly between +4 mm and –18 mm.

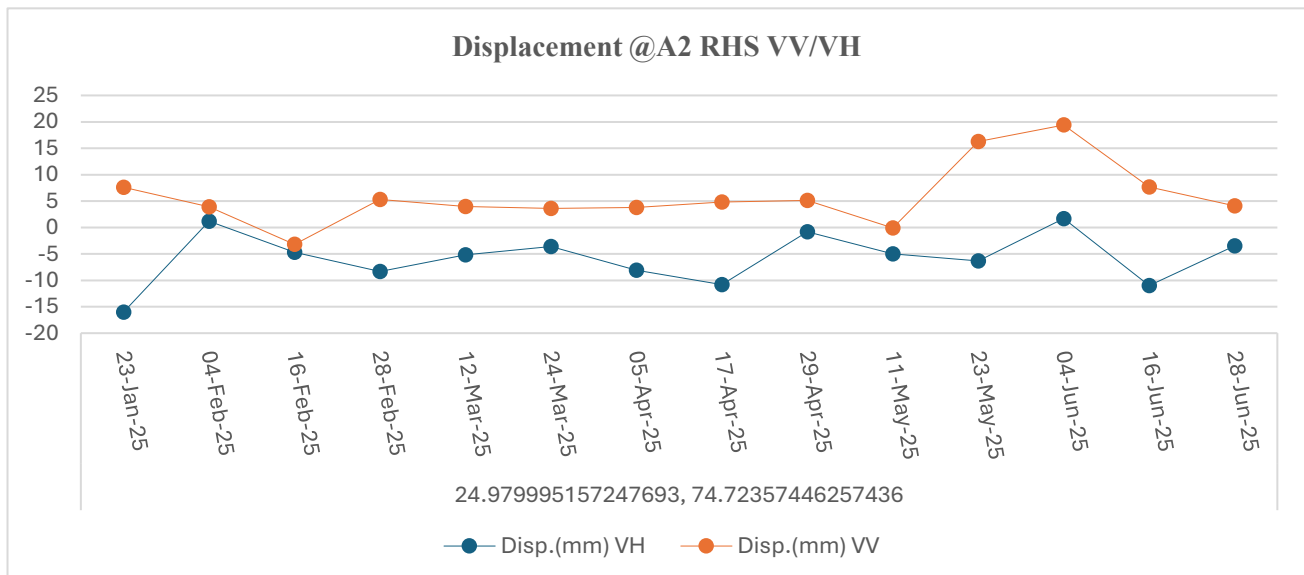
Critical Events & Anomalies:

- **February 2025:** VH rises to +4 mm, while VV remains negative (–10 mm), marking a decoupled response with lateral uplift under vertical settlement.
- **June 2025:** Both VH and VV reach strongly negative values again (VH –18 mm, VV –15 mm), confirming persistent settlement toward the end of the monitoring period.

3.10 SAR Based Analysis at Abutment 2 RHS:



A2 RHS (24.979995157247693, 74.72357446257436)



Key Observations:

Displacement Magnitude Patterns:

- **Vertical Displacement (VV)** Vertical displacement exhibits moderate variability with an overall positive dominance during the monitoring period (Jan–Jun 2025), indicating net vertical uplift at the A2 RHS location.
- **Horizontal Displacement (VH)** Horizontal displacement shows high variability with a predominantly negative trend, ranging roughly between +3 mm and –15 mm.

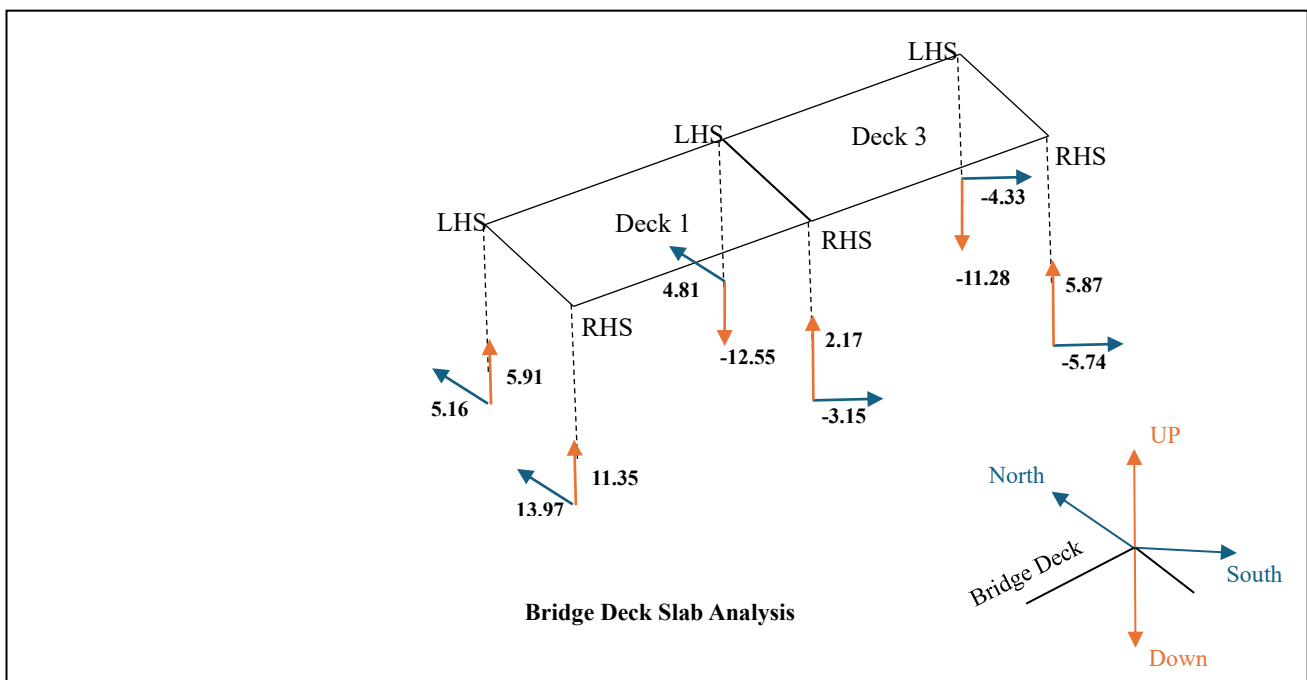
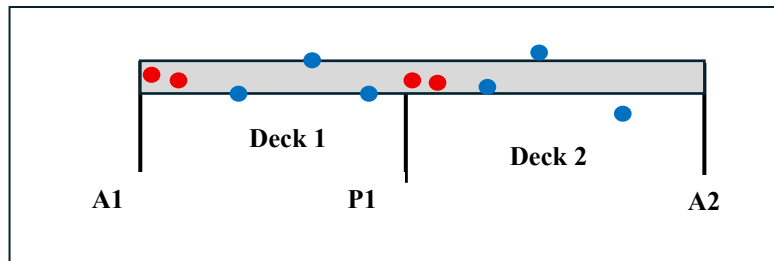
Critical Events & Anomalies:

- **January 2025:** VH begins around –15 mm, while VV is moderately positive (+8 mm), indicating early lateral settlement occurring alongside vertical uplift.
- **April 2025:** VH drops further to –10 mm, while VV increases slightly to +6 mm, indicating a decoupled response with increasing lateral settlement under uplift.
- **June 2025:** VH drops again to –10 mm, while VV reduces to +5 mm, confirming continued lateral settlement with post-peak vertical relaxation toward the end of the monitoring period.

ANALYSIS

1. Identify potential deformation, shifts or movement of the bridge structure

- Points of Analysis (POA) of bridge (●)



Vertical Displacement Behaviour:

- The bridge deck slab exhibits highly non-uniform vertical displacement along its longitudinal extent, indicating differential vertical response across Decks 1 to 3. The alternating uplift–settlement pattern across successive decks confirms combined longitudinal bending and torsional action, rather than uniform vertical sagging.

Horizontal Displacement Behaviour:

- Horizontal displacement shows clear directional variability across both deck width and length, indicating torsion-dominated deformation behaviour. The presence of opposing horizontal movements across decks confirms distributed torsional rotation, rather than rigid-body translation.

CONCLUSION:

- The bridge deck slab demonstrates a complex deformation response, characterized by simultaneous vertical uplift, significant settlement, and asymmetric horizontal displacement across Decks 1–3. Overall, the magnitude and variability of the observed displacements suggest structurally significant behaviour, warranting continued monitoring and detailed correlation with loading configuration, bearing behaviour, and environmental effects.